Timothy Carroll

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Education

University of California San Diego Sep. 2025 - June 2028

Bachelor of Science in Electrical Engineering

- Relevant Coursework:
  - Digital Design & RTL: Advanced Digital Design Project (ECE 111 - In Progress), Active Circuit Design (ECE 102), Linear Electronic Systems (ECE 100), Circuits Lab (ECE 65), Introduction to Digital Design (ECE 25)
  - RF & Semiconductors: Antenna Systems (ECE 123 - Planned), Transmission Lines (ECE 107 - Planned), Semiconductor Devices (ECE 103 - Planned)

College of Marin Aug. 2023 – May 2025

- 3.8 GPA, Honor Roll 2023 - 2025
- Member of the Engineering Club, Vice President of the Physics Club

Experience & Research

iPower3ES Lab (UCSD) – Undergraduate Research Apprentice Oct 2026 - Present

University of California San Diego *La Jolla, CA*

- Analyze emerging methodologies in high-frequency power management systems, miniaturized power conversion, and power integrity architectures during technical briefings.
- Evaluate literature on Power Supply on Chip (PwrSoC) and integrated circuit (IC) designs to identify noise-reduction frameworks critical for sensitive hardware applications.
- Collaborate with graduate mentors to scope a research framework investigating integrated power architectures.

Nanofabrication Research Intern May 2024 - Aug. 2024

Carnegie Mellon University *Pittsburgh, PA*

- Researched semiconductor lithographic and oxide etch processes to optimize resolution and yield; developed high-selectivity oxide etch and lift-off frameworks.
- Operated e-beam lithography, sputtering systems, profilometers, and nanospectrometers in a cleanroom environment while analyzing resist thickness variations.
- Designed a centralized spin-table data system adopted lab-wide and presented research findings to academic mentors and industry professionals.

Broadcast Engineering Intern June 2025 - August 2025

KNTV NBC Bay Area *San Jose, CA*

- Troubleshot ongoing, high-risk points of failure for live Telemundo and NBC broadcasts averaging up to 2.25 million concurrent viewers and generated workflow improvements that minimized air-time disruptions overseen by senior engineering staff.

Founding MESA Program Tutor and Peer Educator Aug. 2024 - May 2025

College of Marin *Kentfield, CA*

- Founded and led a weekly Physics Study Group and provided individual and group tutoring across the full physics sequence; received Employee of the Month and Service to Community awards.

Audio Design Intern Dec. 2023 - Jan. 2024

Roswell Pro Audio *Santa Rosa, CA*

- Evaluated hardware layouts and diagnosed low-level AC noise in professional audio hardware using oscilloscopes, spectrum analyzers, and high-voltage signal-path testing.

Projects

LoRa Telemetry Communications Module - *Triton Astronautics Research Systems* Oct. 2026

- Designing a LoRa-based telemetry subsystem using the SX1276 transceiver and ESP32 flight computer, creating PCB-level interfaces for SPI communications, power regulation, and signal-integrity optimized RF antenna integration.
- Developing RF link-budget calculations, including free-space path loss and predicted received power, to validate communication ranges against a strict minimum 10 dB link-margin requirement.

SystemVerilog RTL Digital Design Portfolio — ECE 111 Sep. 2026 - Dec. 2026

- Developing a hardware accelerator for the SHA-256 cryptographic algorithm in SystemVerilog, optimizing the microarchitecture for area-constrained and delay-optimized RTL design targets.
- Engineered a suite of verified digital blocks including a Parameterized ALU with Registered Output, shift-add multipliers, lookahead carry adders, and Finite State Machines (FSMs).

Technical Skills

Hardware: Oscilloscope, Signal Generator, Multimeter, Soldering & Prototyping, DC/AC Circuit Analysis, Filter Design, Combinational & Sequential Logic

Software/Simulation: C/C++, MATLAB, Verilog (RTL), SystemVerilog, LTspice, KiCad (PCB Design), Quartus Prime, Xilinx Vivado